

Lecture 4:
The Economics of Pollution Control (II)
ECON7720: Ecological and Environmental Economics

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Semester 2, 2026 | Required reading: Tietenberg & Lewis, Ch. 7

Recap of Lecture 3

- **Pollutant taxonomy**: stock vs fund (persistence), uniform vs non-uniform mixing (location matters), guides instrument design.
- **MD** rises with emissions; **MAC** falls. The **efficient** level minimises damage + control cost at $MD(E^*) = MAC(E^*)$, generally **not zero**. The TSC curve is **U-shaped**.
- **Cost-effectiveness**: hit a target at least cost by equalising MAC across sources (**equimarginal principle**, $MAC_i = \lambda$).

Lecture 4: The Economics of Pollution Control (II)

1. The **policy instruments**: command-and-control, emission taxes, abatement subsidies, product charges, and cap-and-trade.
2. Choosing between **price** and **quantity** instruments (revenue, distribution, entry, inflation, technical change).
3. Instrument choice under **uncertainty**: Weitzman's prices vs quantities.
4. **Non-uniformly mixed** pollutants: when location matters.

Learning Resources

Required readings

- Tietenberg & Lewis, Ch. 7 (*Economics of Pollution Control: An Overview*): read it and attempt the self-test exercises.

Recommended readings

- Schmalensee, R. & Stavins, R.N. (2017), "Lessons Learned from Three Decades of Experience with Cap and Trade," *Review of Environmental Economics and Policy* 11(1), 59–79.
- Goulder, L.H. (2013), "Markets for Pollution Allowances: What Are the (New) Lessons?" *Journal of Economic Perspectives* 27(1), 87–102.

Additional material (interactive simulations)

- Standards, taxes, cap-and-trade, Weitzman
- Non-uniform pollutants (location, ambient charges, hotspots)

The Policy Instruments

A menu of instruments

Three broad families

1. **Command-and-control (standards)**: the regulator mandates an emission limit or a technology, firm by firm.
2. **Price instruments (emission taxes/charges)**: put a *price* on each unit of pollution and let firms choose how much to emit.
3. **Quantity instruments (cap-and-trade)**: fix the total *quantity* of pollution and let a market price the permits.

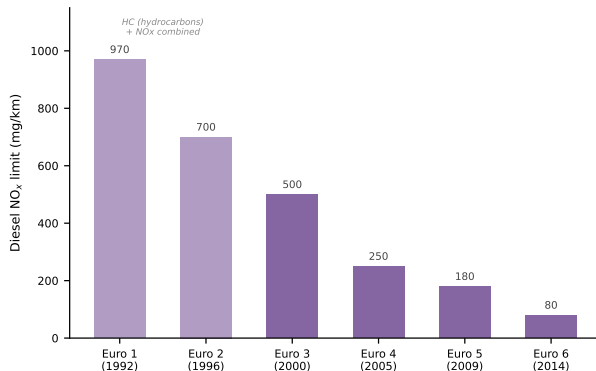
The key question for each: does it reach the target at **least cost** (equimarginal), and what **information** does the regulator need?

(1) Command-and-control — Euro emission standards

Euro emission standards (EU, 1992–): every new car must meet progressively tighter limits on NO_x (nitrogen oxides), PM (particulate matter), and CO (carbon monoxide).

Diesel NO_x : **80 mg/km** at Euro 6, down from 500 at Euro 3.

Effective, but **costly**: the regulator picks the technology, not the firm. Could the same result have been reached at **lower cost**?



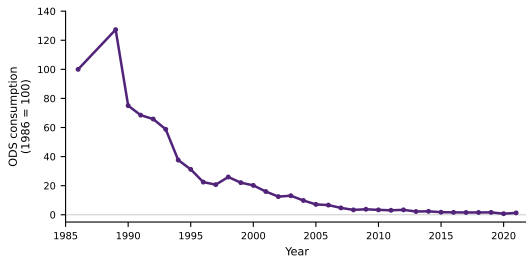
Diesel NO_x limits by Euro standard. Data: EU regulations.

(1) Command-and-control — Montreal Protocol

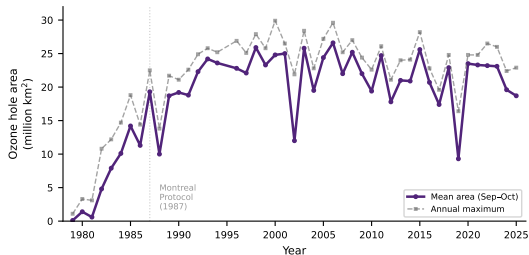
Montreal Protocol (1987): global phase-out of CFCs (chlorofluorocarbons) and halons, ozone-depleting substances.

Widely judged the **most successful** environmental treaty: CFC production virtually eliminated.

Why CAC worked here: substitutes existed, few producers, **negotiation** feasible.



Global ODS consumption indexed to 1986 = 100. Data: UNEP (2023) via Our World in Data.



Antarctic ozone hole area (mean & max). Data: NASA Ozone Watch via Our World in Data.

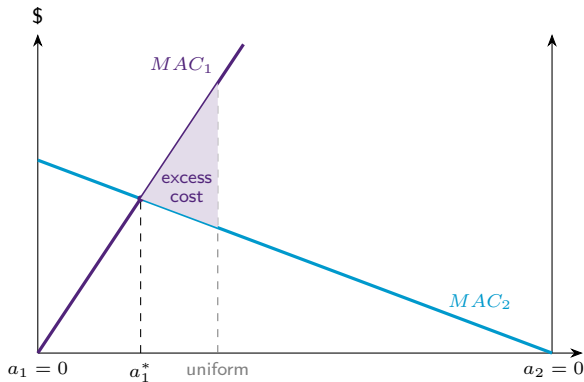
(1) Command-and-control — two common forms of standards

- **Emission (performance) standard**: a legal cap on what each source may emit (e.g. “no more than X tonnes”).
- **Technology (design) standard**: mandate a specific abatement technology (e.g. “install a scrubber or a filter”).

Why standards struggle on cost

A **uniform** standard equalises *emissions*, not *marginal costs*, so in general $MAC_i \neq MAC_j$, not cost-effective. An **individualised** standard *could* be cost-effective, but only if the regulator **knows every firm's MAC**, information firms have no reason to reveal truthfully. A technology standard also **freezes** innovation at the mandated method.

(1) Command-and-control — the excess cost of a uniform standard



A uniform standard forces the **high-cost** firm to over-abate, while the **cheap** firm under-abates.

- Moving from the **equimarginal** split (a_1^*) to equal cuts pushes firm 1 up its steep MAC.
- The **shaded** triangle between the two MAC curves is the **excess cost** of ignoring cost differences.
- Trade (or a tax) recovers exactly this triangle.

(1) Command-and-control — the upside, and where standards fit

- **Simple** to write and, often, to monitor; familiar to courts and regulatory agencies.
- Give **quantity certainty** at each source, attractive for **hotspots** (locations where pollution concentrates to dangerous levels) and acutely toxic substances where any local exposure is dangerous.
- Sometimes the only feasible option when **measuring** actual emissions is impractical; the regulator mandates a specific **technology** instead (e.g. catalytic converters).

The complaint is **not** that standards fail to clean up, they can, but that they do so at **higher cost than necessary** and **reduce the incentive** to innovate. Hence the case for **incentive-based** instruments.

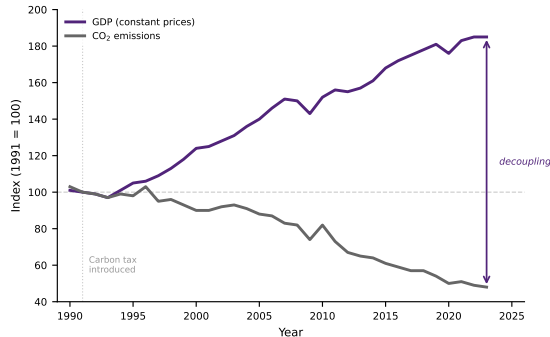
(2) Price instruments — Sweden's carbon tax

Sweden's carbon tax (1991–): one of the world's highest, currently $> \text{€}100/\text{tCO}_2$.

Emissions -30% , GDP $+80\%$ $\Rightarrow$ decoupling.

Revenue-neutral: tax revenue used to cut labour and income taxes, making the reform politically durable.

A price instrument sets a price and lets agents decide *how much* to adjust.



GDP & CO₂ indexed to 1991 = 100. Data: World Bank; Our World in Data.

(2) Price instruments — emission taxes

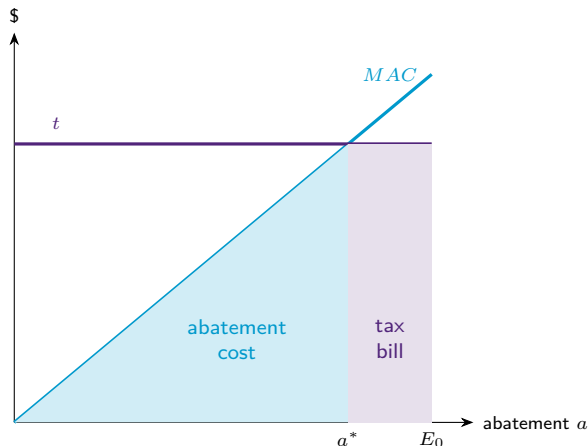
Charge each source a tax t for every unit it emits. A cost-minimising firm keeps abating as long as abating is cheaper than paying the tax:

abate while $MAC < t$, stop where $MAC_i = t$.

- Every firm faces the **same** price t , so every firm sets its MAC_i to the **same** number $\Rightarrow MAC_1 = \dots = MAC_n = t$: the **equimarginal principle holds automatically**.
- The regulator needs **no** firm-specific cost information: the market sorts out who abates.

This is related with the **Pigouvian** idea from Lecture 2: set the price equal to marginal damage and the externality is internalised. Set $t = MD(E^*)$ to deliver the *efficient* level.

(2) Price instruments — the firm's choice under a tax



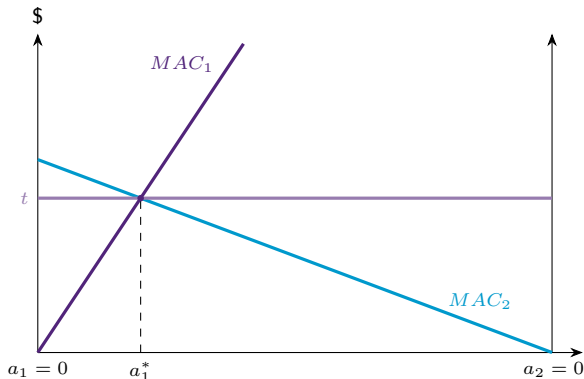
Abate to $MAC = t$

The firm cleans up while the next unit is cheaper than the tax, stopping at a^* .

- **Abatement cost** $= \int_0^{a^*} MAC$: the real resource cost of cleaning up.
- **Tax bill** $= t \times (E_0 - a^*)$: paid on the *remaining* emissions.

The firm bears *both*; society bears only the abatement cost.

(2) Price instruments — a tax with multiple firms



Cost-effectiveness

Every firm abates to $MAC_i = t$:

- Firm 1 (steep MAC): abates **less**; cleaning up is expensive.
- Firm 2 (flat MAC): abates **more**; cleaning up is cheap.

The tax **automatically** reaches the equimarginal allocation, the same point that trade reaches.

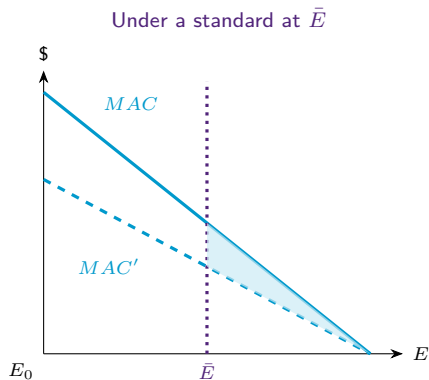
No central planner needs to know each firm's MAC; the price signal does the work.

(2) Price instruments — taxes sharpen the incentive to innovate

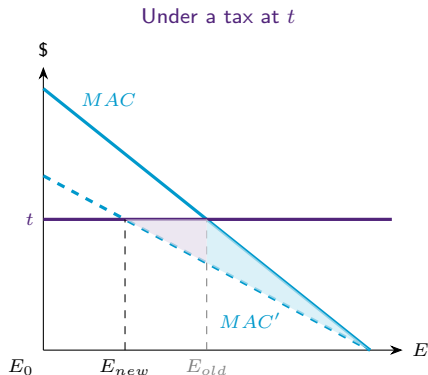
Under a **standard**, once a firm meets the limit it has **no reason** to clean up further. Under a **tax**, every unit still emitted keeps costing t , so the firm has a **standing incentive** to find cheaper abatement and cut more.

- New technology **lowers MAC** $\Rightarrow$ the firm abates **more** *and* shrinks its tax bill: it saves on **two** margins.
- Under a standard it saves only the control cost of meeting a **fixed** limit: a **weaker** reward.

(2) Price instruments — cost savings from technological change



Firm still emits $\bar{E}$. Shaded area = saving: reduced control cost of meeting the same limit.



Firm abates more. Shaded area = saving on old units; shaded area = net gain from extra abatement (saved tax — extra abatement cost).

Adapted from Tietenberg & Lewis (2024), Figure 7.5.

(2) Price instruments — abatement subsidies (static mirror)

Instead of *charging* $\$t$ per unit emitted, *pay* $\$s$ for every unit *abated*. In the short run, with a fixed number of firms, the marginal calculus is the *same*:

$$\text{abate while } MAC < s, \quad \text{stop at } MAC_i = s.$$

So a subsidy $s = t$ delivers the *same* cost-effective allocation as the tax *for existing firms*. The difference is *who pays*: the tax takes money *from* firms; the subsidy pays money *to* them (a budget cost for government). That distributional flip has a sting in the tail.

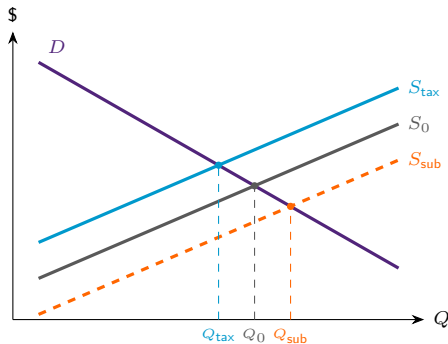
(2) Price instruments — abatement subsidies (entry problem)

A subsidy raises a firm's **profit** (it is paid for abating), so:

- more firms **enter** (or fewer exit) the industry;
- with more firms, **total** emissions can **rise**, even though each firm abates “correctly” at the margin.

A **tax** does the reverse: it lowers profit, encourages **exit**, and shrinks total emissions. So tax and subsidy are **not** equivalent once entry/exit is allowed.

(2) Price instruments — abatement subsidies (why total emissions can rise)



A **tax** shifts supply **up** (S_{tax}): fewer firms, less output, **fewer** total emissions.

An abatement **subsidy** shifts supply **down** (S_{sub}): more firms enter, more output, total emissions can **rise**, even though each firm abates at the margin.

$$Q_{\text{tax}} < Q_0 < Q_{\text{sub}}$$

The **per-firm** abatement is identical, but the **number of firms** differs. This is why tax and subsidy diverge in the **long run**.

(2) Price instruments — other instruments in the toolkit

- **Deposit–refund** (container deposits, lead-acid batteries, e-waste): a tax at purchase *refunded* on proper return; internalises disposal without monitoring every bin.
- **Feebates**: fees on dirty products fund rebates on clean ones; revenue-neutral by design.
- **Product charges**: when the *emission* itself is hard to monitor, tax the *product* whose use causes the pollution. Examples: a plastic-bag charge. Most efficient when each unit of the product causes roughly the same marginal damage.
- **Liability and bonds**: assurance bonds and clean-up liability make firms internalise *future* damage.
- **Information**: disclosure, eco-labels, pollutant registers; cheap, and surprisingly effective where reputations matter.

Real policy depends on what can be **measured**, **monitored**, and **enforced**.

(2) Price instruments — example: liability and bonds in mining



Brumadinho tailings dam collapse (Vale, Jan 2019). 270 killed; clean-up cost >US\$7 bn. Photo: Ibama, CC BY-SA 2.0.

Mining **tailings** (slurries of rock, water, chemicals) can cause **billions** in damage if a dam fails.

- **Assurance bond**: deposit *before* operations; refunded only on **certified** rehabilitation. Shifts future damage into **present** decisions.
- **Strict liability**: operator liable for *all* damage regardless of fault, continuous incentive to reduce risk.
- **Financial provisioning**: funds set aside **progressively** as disturbed area grows (e.g. Queensland FPS, 2019).

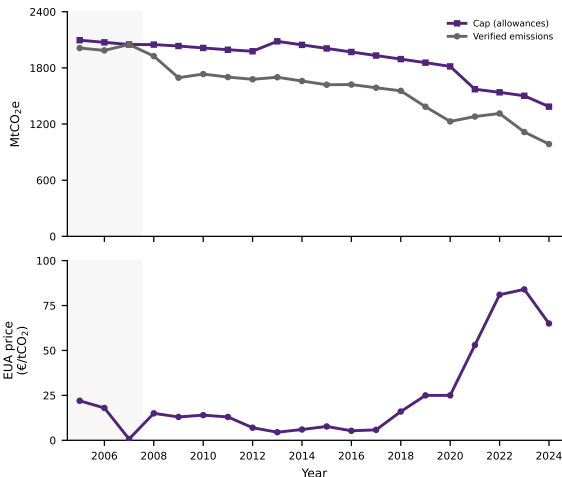
Standards say *how* to build a dam; bonds and liability ask *who pays if it fails*.

(3) Cap-and-trade — example

EU ETS (2005–): ~10 000 installations
(power, industry, aviation).

Cap declines annually: 1.74%/yr
(Phase III) → 2.2%/yr (Phase IV).

Early **over-allocation** crashed prices.
Reforms (auctioning, Market Stability
Reserve) tightened supply ⇒ prices above
€80/tCO₂.



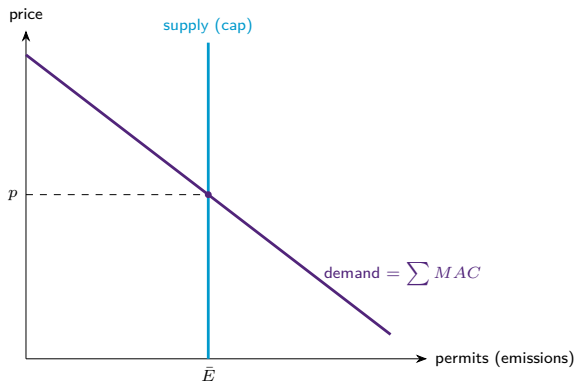
Cap & emissions: EEA EU ETS Data Viewer; EC COM(2025)735. Price: Ember Carbon Price Tracker; ICAP.

(3) Cap-and-trade — the idea

1. **The cap:** government issues a fixed number of **permits** (= total allowed emissions $\bar{E}$). The environmental target is set *by construction*.
2. **Allocation:** hand the permits out **free** (“free allocation”) or **auction** them.
3. **Trade:** permits are freely **tradeable**. A firm must hold one permit per unit emitted.

A firm with **high** MAC would rather **buy** a permit than abate; a firm with **low** MAC would rather abate and **sell** its permit. Trade flows until no such gains remain.

(3) Cap-and-trade — the permit market



A vertical supply

The cap fixes **supply** at $\bar{E}$ (a vertical line). **Demand** to hold permits is the aggregate willingness to pay to emit; the **aggregate MAC**.

Their crossing sets the **equilibrium permit price** p .

Tighten the cap (shift supply left) $\Rightarrow$ the price **rises**. The regulator picks the **quantity**; the market reveals the **price**.

(3) Cap-and-trade — the equilibrium price

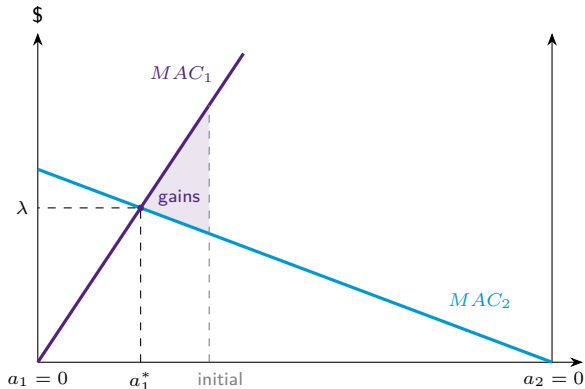
A firm trades permits until its MAC equals the market permit price p :

buy while $MAC > p$, sell while $MAC < p$, stop at $MAC_i = p$.

- In equilibrium all firms share the same $MAC = p$: the equimarginal principle again holds automatically.
- The permit price p settles at exactly the MAC^* that clears the cap. The permit price p equals the tax t that would achieve the same target.

Tax and cap are duals: a tax fixes the price and lets quantity adjust; a cap fixes the quantity and lets the price adjust. If the regulator knows the MD and MAC curves with certainty, the tax that delivers E^* and the cap set at E^* produce the same outcome. They diverge only under uncertainty (Weitzman, see later).

(3) Cap-and-trade — the gains from trade



Equal free allocation gives each firm the same permits. At that split $MAC_1 \gg MAC_2$.

- The high-cost firm **buys** permits; the low-cost firm **sells** and abates more.
- Trade continues until $MAC_1 = MAC_2 = \lambda$.
- The **shaded** area is the cost saving from trade.

(3) Cap-and-trade — the independence (Coase) property

Whoever receives the permits, trade moves them to the **same** cost-effective allocation and the **same** price.

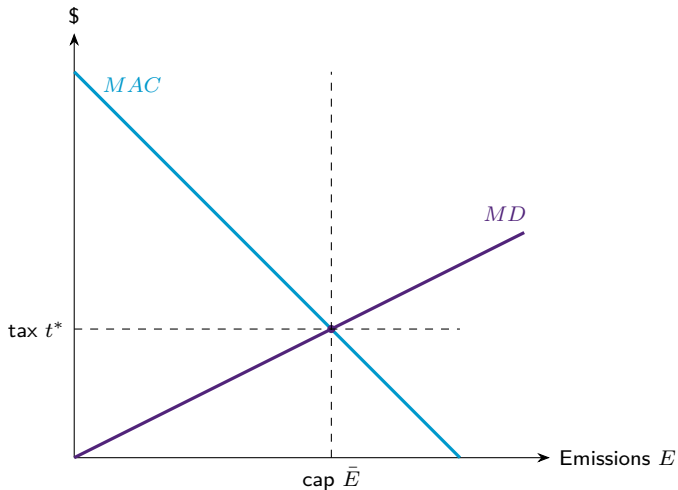
- Free permits are a **lump-sum transfer**: a firm that receives more can sell the surplus and is **richer**, but it still abates until $MAC = p$. The marginal calculation (“is it cheaper to clean up or to hold a permit?”) does not depend on how many permits were given for free.
- So the **distribution** of permits sets **who gains**, while the **market** sets **who abates**.

This is the cap-and-trade analogue of Lecture 2's Coase result: with low transaction costs, the efficient outcome is reached regardless of the initial assignment of rights, only the **distribution** differs.

(3) Cap-and-trade — caveats

- **Market power:** if one firm dominates the permit market it can **withhold** permits to raise rivals' costs, the price no longer equals the true marginal cost.
- **Transaction costs:** search, broker fees and approval delays can choke **thin** markets, so cheap trades go unmade.
- **Monitoring, reporting, and verification:** a price on pollution only works if emissions can be **measured** and **enforced**; pollutants that are easy to monitor (e.g. CO₂ from fuel use) suit market instruments better than those that are hard to track (e.g. agricultural runoff).
- **Hotspots:** with *non-uniform* pollutants, trading can concentrate emissions where they do most harm.

(3) Cap-and-trade — tax and cap, two routes to the same point



A **tax** fixes the *height* (t^*); a **cap** fixes the *width* ($\bar{E}$). If the regulator knows MD and MAC with certainty, both meet at $MD = MAC$: the instruments are **duals**. They diverge under **uncertainty** (next section).

(3) Cap-and-trade — banking and borrowing permits

Most modern schemes let firms **bank** unused permits for later (and sometimes **borrow** against future allocations).

- **Banking** rewards *early* abatement and **smooths** the permit price when costs or demand fluctuate.
- It lets firms shift cuts to when they are **cheapest**, lowering total cost, an intertemporal version of the equimarginal principle.
- **Caveat**: a large bank can hold the price down for years, weakening the near-term signal.

(3) Cap-and-trade — how are permits handed out?

1. **Free allocation:** permits given away, no revenue raised:
 - **Grandfathering:** based on *historical* emissions. Builds political support, but rewards past polluters.
 - **Benchmarking:** based on current *output* $\times$ best-practice rate. Rewards clean producers; used for trade-exposed sectors.
2. **Auctioning:** firms **buy** permits. Raises revenue, reveals the price, treats entrants and incumbents alike.

By the **independence property** all methods reach the same cost-effective allocation; they differ in **distribution** and **revenue**. The trend is *away* from free allocation toward auctioning.

(3) Cap-and-trade — baseline-and-credit schemes

Instead of a fixed **cap** on total tonnes, set each facility a **baseline** (often an emissions-*intensity* rate $\times$ its output). Facilities below baseline earn **credits**; those above must **buy** them.

- Prices emissions **above** the baseline, like a tax on the excess.
- **Output-based** baselines blunt leakage and protect competitiveness. . .
- . . . but because the baseline **scales with output**, **total** emissions are *not* hard-capped, they can rise if the sector grows.

Australia's **Safeguard Mechanism** (reformed 2023) is exactly this design: ~ 215 large facilities each receive an output-based emissions baseline that declines $\sim 4.9\%$ per year to 2030 (consistent with Australia's 43% reduction target). Facilities below their baseline earn **Safeguard Mechanism Credits** (SMCs); those above must surrender SMCs or **Australian Carbon Credit Units** (ACCUs).

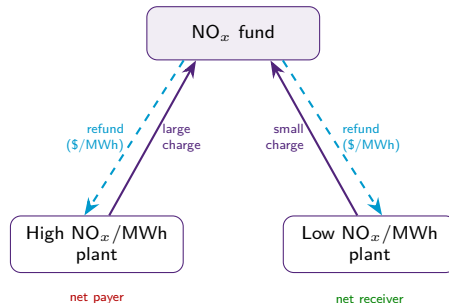
Example — Sweden's NO_x charge (a refunded emission charge)

Since 1992, Sweden charges large combustion plants per kg of NO_x emitted: initially SEK 40/kg, raised to SEK 50/kg in 2008 ($\approx \text{€} 4.5/\text{kg}$).

All revenue is refunded to the same plants, in proportion to their energy produced (MWh of electricity and heat), not their emissions.

- Plants with high NO_x/MWh are net payers.
- Plants with low NO_x/MWh are net receivers.

Result: specific NO_x emissions fell $\sim 50\%$ in the scheme's first decade.



$$\text{Net transfer} = \text{charge} \times (\text{own NO}_x/\text{MWh} - \text{sector average NO}_x/\text{MWh}) \times \text{MWh}.$$

Why command-and-control still dominates

Despite the economist's case, most pollution is still governed by **standards**. Why?

- **Simplicity and familiarity**: easier to legislate, litigate and enforce.
- **Monitoring**: where emissions cannot be metered cheaply, a **technology** mandate is the only workable rule.
- **Distributional politics**: standards hide the cost; an explicit price is visible and easy to attack.
- **Certainty at the source**: vital for **local toxics** and hotspots.

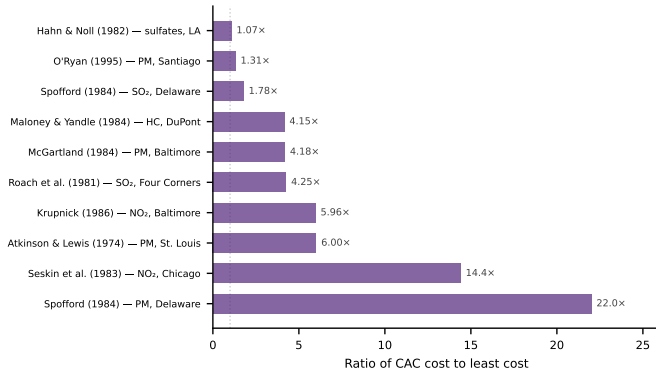
The frontier is **hybrid**: performance standards *with* tradeable credits keep some flexibility while preserving CAC's familiarity.

Standards vs incentives

Across 10 US air-pollution studies, command-and-control cost **1.07 to 22×** the least-cost benchmark for the *same* environmental result.

The wider the bar, the larger the **excess cost** of ignoring firms' heterogeneous abatement costs.

Realised market programs (SO₂, lead) delivered **large savings**, though usually less than the theoretical maximum.



Data: Table 15.2 in Tietenberg & Lewis (2024), compiled from 10 US air-pollution studies.

Choosing an instrument

Questions to ask before reaching for a tool

1. Is the pollutant **uniformly mixed**, or does **location** matter?
2. Can emissions be **measured** and **enforced** cheaply? If not, lean on technology standards.
3. Is **quantity** certainty critical (thresholds, hotspots), or is **cost** certainty more valuable?
4. How important is **revenue**, and who should bear the cost?
5. How **competitive** and **thick** will any permit market be?
6. What is **politically** durable?

What information does the regulator need?

Instrument	Cost-effective?	Needs each firm's MAC?
Uniform standard	No	No
Individual standard	Yes	Yes (firm by firm)
Emission tax	Yes	No
Cap-and-trade	Yes	No

The **decisive advantage** of price and quantity instruments: they reach the equimarginal allocation **without** the cost information that firms would never truthfully reveal. The individual standard *can* match them, but only if the regulator somehow **knows every MAC**.

Instruments criteria

Criteria	Standard	Emission tax	Cap-and-trade
Cost-effective?	only if individual	yes	yes
Needs firms' MAC?	yes (individual)	no	no
Certain <i>quantity</i> ?	at each source	no	yes
Certain <i>price</i> ?	no	yes	no
Raises revenue?	no	yes	if auctioned
Innovation incentive	weak	strong	strong

The *quantity*-vs-*price* rows are where tax and cap part company.

Price vs Quantity Instruments

Price vs quantity instruments — when do they diverge?

We compare them on the dimensions where the equivalence breaks:

1. **Revenue**: who collects the rents?
2. **Distribution**: who ultimately bears the cost?
3. **Entry** of new emitters: what happens to total pollution?
4. **Inflation**: does the instrument hold its value?
5. **Technical progress**: who captures the gains?
6. **Uncertainty** about the MAC and MD curves (Weitzman).

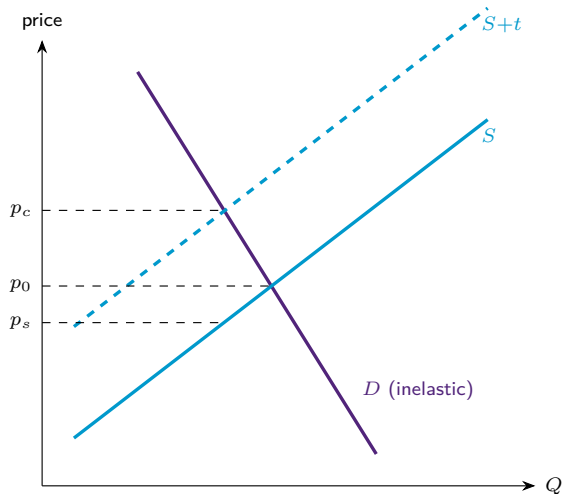
Dimension 1 — revenue

Which instruments raise money?

Instrument	Raises public revenue?
Emission tax	Yes
Auctioned permits	Yes
Freely allocated permits	No (rents go to firms)
Standards	No

Revenue is valuable: it lets the government **cut a distortionary tax** (on labour or capital) using the pollution revenue, improving the environment *and* the tax system.

Dimension 2 — who really pays? (incidence)



The legal payer is not the bearer

A tax on emissions raises product prices.

Who bears it depends on elasticities, not on who writes the cheque. With inelastic

demand (petrol, electricity), consumers bear most: p_c rises nearly by t . Same logic applies to the permit price firms pass through.

Dimension 3 — entry of new emitters

The environmental difference

- Under a **tax**: each new firm emits up to where $MAC = t$. More firms $\Rightarrow$ **more total emissions**. The price is fixed; the quantity floats **up**.
- Under a **cap**: the total is **fixed** by the number of permits. A new entrant must **buy** permits from incumbents, bidding the **price up** but leaving **total emissions unchanged**.

Dimension 4 — inflation

Does the instrument hold its real value?

- A **tax** fixed in nominal dollars is **eroded** by inflation: firms abate **less**, emissions **rise**, unless the tax is indexed.
- A **cap** is denominated in **physical units** (tonnes). The permit **price** rises with the general price level, but the **quantity** of pollution is unaffected.

Same theme as entry: the **quantity instrument** automatically protects the *environmental* target; the **price instrument** protects the *cost* but lets quantity slip.

Dimension 5 — technological progress

When MAC falls industry-wide

- Under a **tax**: cheaper abatement $\Rightarrow$ firms abate **more** at the same price $\Rightarrow$ total emissions fall **below** the old level. **The environment gains.**
- Under a **cap**: total emissions are **pinned at the cap**; cheaper abatement just **lowers the permit price**. **Firms gain**, the environment does **not**, unless the regulator **tightens the cap**.

Role reversal: now the **price** instrument delivers the extra *environmental* bonus. With a fixed cap, innovation becomes cost savings. Hence real cap-and-trade systems **tighten the cap** over time.

Dimension 6 — uncertainty: prices vs quantities (Weitzman 1974)

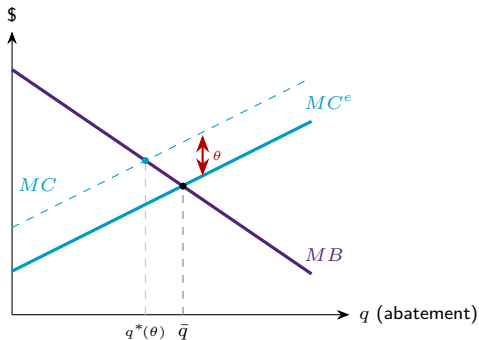
The regulator only *estimates* the curves

Set the tax t or the cap $\bar{E}$ using a *guess* of the MAC curve. If the true MAC differs, the two instruments make *different errors*:

- A *tax* pins down the *cost* per unit but lets the realised *quantity* miss the target.
- A *cap* pins down the *quantity* but lets the realised *cost* (permit price) be wrong.

Which error is worse depends on the relative slopes of MD and MAC. Let us make that precise.

Weitzman — a linear-quadratic set-up



Timing: (1) regulator sets t or $\bar{q}$ using MC^e ($\theta=0$); (2) nature draws θ ; (3) firms respond given true MC . If θ were known, no welfare loss.

Work in **abatement** q :

$$MB(q) = \beta - bq$$

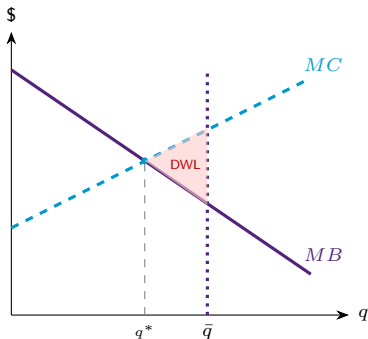
$$MC(q; \theta) = \gamma + cq + \theta$$

θ : **cost shock** (mean 0, variance σ^2).

Ex-post optimum

$$q^*(\theta) = \frac{\beta - \gamma - \theta}{b + c}$$

Weitzman — the loss under a quantity



Fix $\bar{q} = q^*(0)$ (best guess). The shock θ moves the true optimum to $q^*(\theta)$, but quantity is **stuck** at $\bar{q}$.

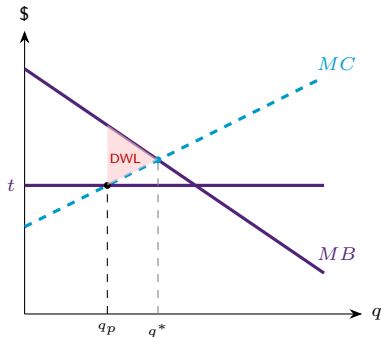
The **shaded triangle** is the deadweight loss: between q^* and $\bar{q}$, the realised MC exceeds MB .

$$\text{Loss}_Q = \frac{\theta^2}{2(b+c)}$$

$$\mathbb{E}[\text{Loss}_Q] = \frac{\sigma^2}{2(b+c)}$$

A **steep** MB (large b) shrinks the triangle, “steep MD $\Rightarrow$ quantity”.

Weitzman — the loss under a price



Fix $t = MB(\bar{q})$ (expected marginal cost). Firms adjust to the shock: $MC = t \Rightarrow q_p = \bar{q} - \theta/c$.

The **shaded triangle**: between q_p and q^* , the MB of abatement exceeds the tax; abatement that **should** happen but doesn't.

$$\text{Loss}_P = \frac{b^2 \theta^2}{2c^2(b+c)}$$

$$\mathbb{E}[\text{Loss}_P] = \frac{b^2 \sigma^2}{2c^2(b+c)}$$

The price lets q **adjust**, good when c is large (small adjustment).

Weitzman — the comparative-advantage formula

Subtract the two expected losses:

$$\Delta = \mathbb{E}[\text{Loss}_Q] - \mathbb{E}[\text{Loss}_P] = \frac{\sigma^2}{2(b+c)} \left(1 - \frac{b^2}{c^2}\right) = \frac{\sigma^2}{2c^2} (c - b) .$$

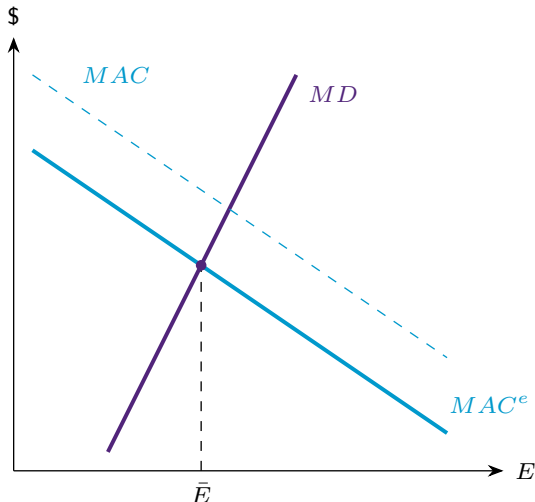
The rule

$\Delta > 0 \iff c > b$ (MAC steeper than MD) $\Rightarrow$ **prices**,

$\Delta < 0 \iff b > c$ (MD steeper than MAC) $\Rightarrow$ **quantities**.

Only the **relative slopes** matter; the size of σ^2 scales *how much* the choice is worth, not *which* wins.

Weitzman — steep marginal damage $\Rightarrow$ use a quantity



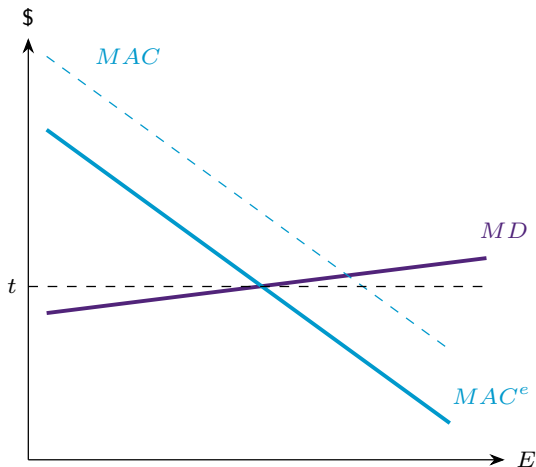
Get the quantity right

When MD is steep (damage shoots up if pollution overshoots), being wrong on *quantity* is very costly. A **cap** holds $E = \bar{E}$ whatever

MAC turns out to be, capping the damage. A tax would let E swing far when MAC surprises.

Threshold / tipping-point pollutants $\Rightarrow$ favour the **cap**.

Weitzman — flat marginal damage $\Rightarrow$ use a price



Get the price right

When MD is flat (damage barely changes with a bit more or less pollution), a *quantity* miss is cheap, but forcing the wrong *cost* on firms is not. A **tax** holds marginal control cost at t whatever MAC turns out to be, avoiding wildly expensive compliance. The quantity drifts, but cheaply. **Stock pollutants** like CO_2 (flat *annual* MD) $\Rightarrow$ favour the **price**.

Compare the slopes

- MD steeper than MAC ($b > c$) $\Rightarrow$ use a **quantity** (cap-and-trade). Errors in pollution are dangerous; lock the quantity.
- MAC steeper than MD ($c > b$) $\Rightarrow$ use a **price** (tax). Errors in pollution are tolerable; lock the cost.

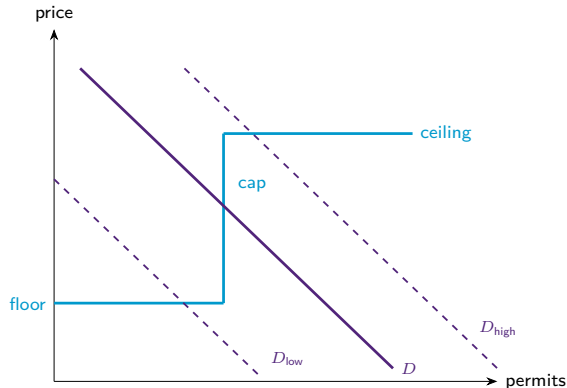
Hybrids

A cap-and-trade with a price collar adds:

- **Price ceiling:** if the permit price rises too high, the regulator **sells** extra permits, capping abatement costs.
- **Price floor:** if the price falls too low, the regulator **withholds** permits, keeping the incentive to abate.

D clears against the cap; D_{high} hits the ceiling; D_{low} hits the floor.

Result: **quantity** certainty in normal times; **cost** certainty in the tails, the system behaves like a **tax** at the extremes.

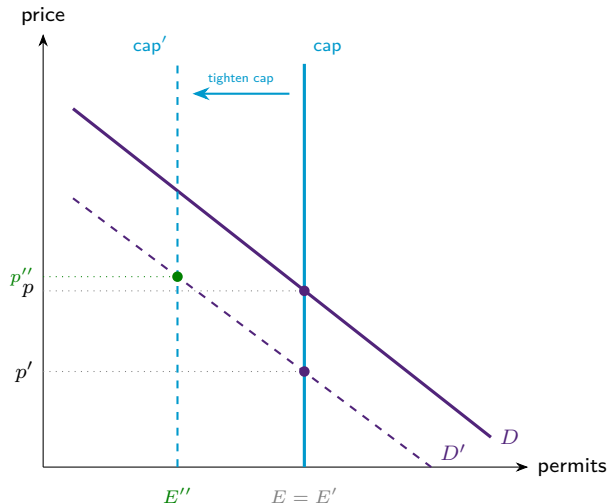


Overlapping policies — the waterbed effect

A renewables subsidy cuts demand for permits ($D \rightarrow D'$), but the cap is **vertical**: price falls, emissions **unchanged**, the **waterbed effect**.

Fix: cancel surplus allowances ($\text{cap} \rightarrow \text{cap}'$)
 $\Rightarrow$ emissions **do** fall, price recovers (p'').

EU ETS evidence: price fell from $\sim \text{€}30$ to $< \text{€}5$ (2008–13); renewables and recession both might have contributed.



When Location Matters

Non-uniformly mixed pollutants

Same tonnage, different harm

For smog precursors, particulates, or wastewater, what matters is the **concentration** people are exposed to at a **receptor** (a monitoring point), and that depends on **where** the source sits.

- A source **upwind** of a city does more harm per tonne than one downwind.
- So “equalise emissions”, and even “equalise MAC ”, is no longer the right cost-effective rule.

We track each source's contribution to concentration via a **transfer coefficient**.

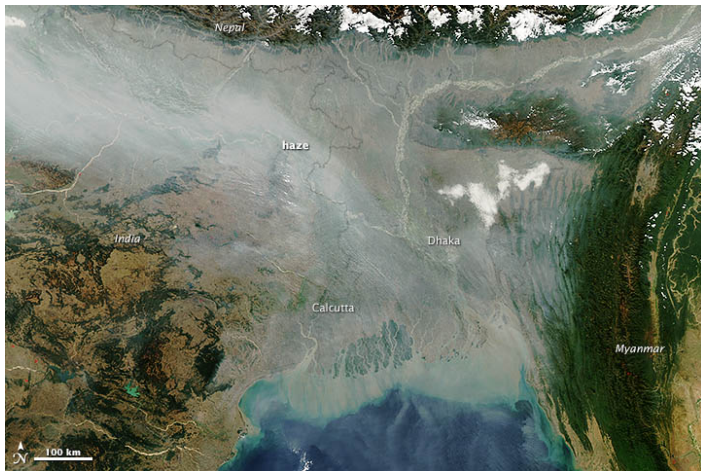
When location matters — the Indo-Gangetic haze

Every winter, crop-burning in Punjab sends a “river of haze” southeast across the Indo-Gangetic Plain.

It blankets Delhi (300 km downwind), Calcutta, and Bangladesh.

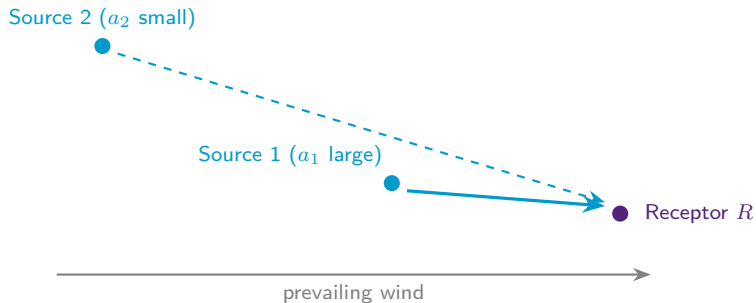
PM_{2.5} in Delhi regularly exceeds WHO limits by 10–20×.

The sources are hundreds of kilometres away.



NASA MODIS, Jan 2013. Public domain.

The geography of the problem



The **same** tonne emitted by Source 1 (close, upwind) raises concentration at R by **more** than a tonne from Source 2 (far). The **transfer coefficient** a_i encodes this. Policy must weight emissions by **where** they come from.

Transfer coefficients and concentration

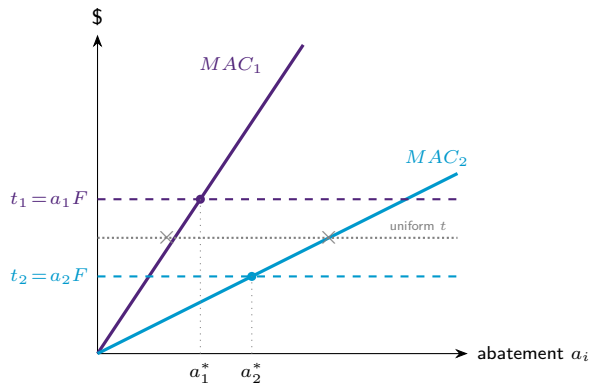
Let K_R be the pollutant concentration at receptor R , and let a_i (the *transfer coefficient*) be the rise in concentration at R per unit of emissions from source i . Then

$$K_R = \sum_{i=1}^I a_i E_i + B,$$

where B is the background concentration (nature, or sources outside the control region).

- A large a_i = a source that hits the receptor hard (close, upwind). Coefficients come from dispersion models (wind, stack height, chemistry).
- The policy target is now a ceiling on K_R (an ambient standard), not on total tonnes.

Cost-effectiveness with location



Equalise MAC_i/a_i , not MAC_i

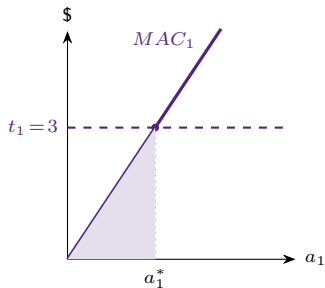
A unit abated at source i lowers K_R by a_i , at cost MAC_i . Least cost:

$$\frac{MAC_1}{a_1} = \frac{MAC_2}{a_2} = F$$

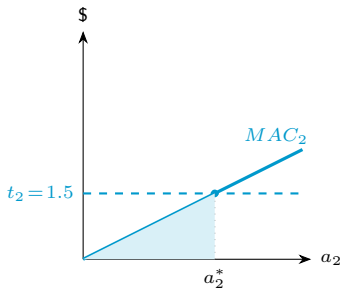
- **Circles:** correct allocation, different MACs but equal MAC/a .
- **Crosses:** uniform tax, equal MACs but **wrong** MAC/a .
- High-impact sources (big a_i) face a **higher** charge and abate less in tonnes, but more per unit of concentration.

Ambient charges

A source-specific price: $t_i = a_i F$



Source 1 (close, $a_1 = 1.0$)



Source 2 (far, $a_2 = 0.5$)

With $F = 3$:

- Source 1 (close): $t_1 = 1.0 \times 3 = 3$ — **high** charge.
- Source 2 (far): $t_2 = 0.5 \times 3 = 1.5$ — **low** charge.

Each firm abates to $MAC_i = t_i$, so $MAC_i/a_i = F$ for both: the **cost-effective** rule.

Contrast a single emission tax (one t for all): it ignores location, over-charges distant sources and under-charges close ones, meeting the ambient standard at **higher** cost.

Ambient permits

With an ambient **charge**, the regulator must know a_i and set each source's fee. Can the **market** do this instead?

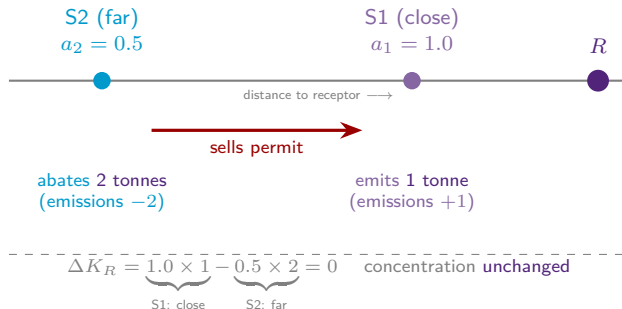
Issue permits that each entitle the holder to raise K_R by a fixed amount ΔK_R . For source i , one permit allows:

$$\Delta E_i = \frac{\Delta K_R}{a_i} \quad \text{tonnes of emission}$$

- A source **close** to the receptor (large a_i) gets **fewer** tonnes per permit.
- A source **far** away (small a_i) gets **more** tonnes per permit.
- All permits trade at a **single price** P , the market automatically weights sources by location.

This is the market analogue of the ambient charge: it achieves the **cost-effective** allocation without the regulator knowing any firm's abatement cost.

Ambient permits — the trading ratio



$S2$'s tonnes only contribute **half** as much concentration at R as $S1$'s ($a_2 = 0.5$ vs $a_1 = 1.0$). So to offset one extra tonne from $S1$, $S2$ must abate **twice as much**: $\Delta E_1 = \frac{a_2}{a_1} \Delta E_2 = \frac{1}{2} \Delta E_2$ (**trading ratio**).

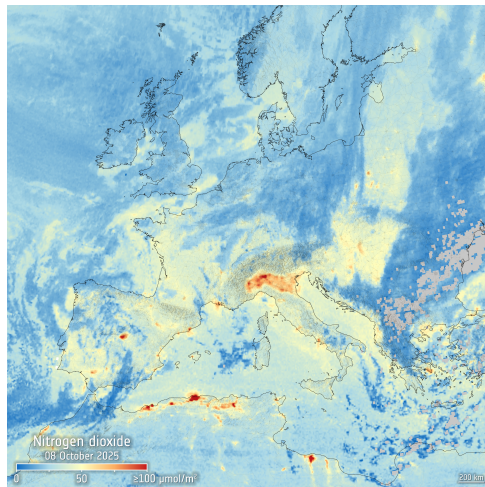
Total emissions fall (-1 tonne), but the **concentration at R** , what matters for damage, is unchanged.

Hotspots — the danger of location-blind trading

For **non-uniform** pollutants (NO_x , $\text{PM}_{2.5}$, SO_2), trading permits **1:1** ignores a_i . Permits migrate to **high-impact** sources, concentrating pollution at one receptor, a **hotspot**, even as total tonnes meet the cap.

- **RECLAIM** ($\text{LA NO}_x/\text{SO}_x$) created local hotspots in already-burdened communities.
- An **environmental-justice** problem: pollution burdens fall on disadvantaged neighbourhoods.
- **Fixes**: trading ratios, zonal restrictions, or local concentration caps.

Note: CO_2 is **uniformly mixed** ($a_n = 1 \forall n$), so the EU ETS has no hotspot problem. Hotspots arise only for **local/regional** pollutants like NO_x and $\text{PM}_{2.5}$.



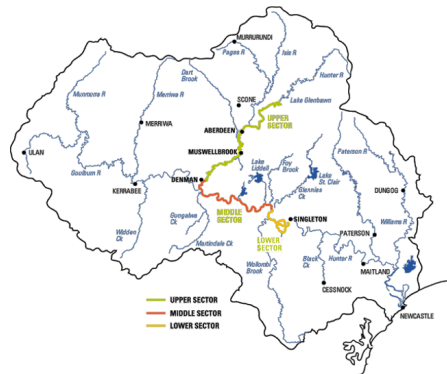
NO_2 over Europe, 8 Oct 2025. The Po Valley hotspot: Alps and Apennines trap emissions in a basin, amplifying a_i for local receptors. Copernicus Sentinel-4, ESA.

Water-quality trading — a non-uniform application

A discharge **upstream** affects every downstream user;
one downstream affects no-one above it.

- **Hunter River Salinity Trading** (NSW, 1995–): mines and power stations hold tradeable salinity credits; may discharge only when river flow is high enough to dilute. Flow- and location-aware.

Water shows the **promise** (big cost savings) and the **difficulty** (monitoring diffuse runoff) of non-uniform trading.



Hunter River Salinity Trading Scheme: upper, middle and lower sectors with location-aware trading. Source: NSW EPA.

In practice — many receptors and imperfect coefficients

With **many** receptors, each source affects *each* monitoring point through its own a_{ij} . The full problem becomes a large constrained optimisation.

The coefficients come from **atmospheric / hydrological dispersion models** (stack height, wind, terrain, chemistry), but they are **estimates** that **change** with weather and season.

The pragmatic fix

- Define **zones**: allow 1:1 trading *within* a zone (treated as uniformly mixed), apply **trading ratios** between zones.
- Accepts a little inefficiency for a **workable** market, chasing a perfectly tuned a_{ij} matrix is neither feasible nor necessary.

Emission permits vs ambient permits

	Emission permit	Ambient permit
Denominated in	tonnes	concentration (ΔK_R)
Trade ratio	1:1	a_j/a_i
Handles location?	no	yes
Hotspot risk	high	low
Admin burden	low	high

For a **uniformly mixed** pollutant the two coincide (equal a_i , so ratios are 1:1). The extra machinery of ambient permits is the **price of location**.

The mathematics of cost-effective pollution control (1/9)

Setup

Consider N pollution sources indexed by $n = 1, \dots, N$.

- Each source emits u_n units in the absence of any control.
- Source n can **abate** q_n units at private cost $C_n(q_n)$, with $C'_n > 0$ and $C''_n > 0$ (increasing marginal cost).
- A **transfer coefficient** a_n captures how much one unit of emission from source n raises the pollutant concentration at a receptor R .

The concentration at the receptor is:

$$K_R = \sum_{n=1}^N a_n(u_n - q_n) + B$$

where B is the background concentration from natural or external sources.

This is fully **general**: it nests uniformly mixed pollutants ($a_n = 1 \forall n$) and non-uniformly mixed pollutants (a_n vary by source location).

The mathematics of cost-effective pollution control (2/9)

The regulator's cost-minimisation problem

The regulator wants to meet an **ambient standard** $K_R \leq \bar{K}$ at the **lowest possible total abatement cost**:

$$\min_{q_1, \dots, q_N} \sum_{n=1}^N C_n(q_n) \quad \text{s.t.} \quad \sum_{n=1}^N a_n(u_n - q_n) + B \leq \bar{K}$$

- The **choice variables** are how much each source abates $(q_1, \dots, q_N)$.
- The **constraint** says that remaining emissions, weighted by their impact on the receptor, must not exceed the standard.
- The regulator does **not** need to know each firm's cost function to *set up* this problem, but solving it *directly* would require that information.

The question is: which policy instrument can **decentralise** this solution without the regulator knowing $C_n(\cdot)$?

The mathematics of cost-effective pollution control (3/9)

Lagrangian

Attach a multiplier $\lambda \geq 0$ to the ambient constraint:

$$\mathcal{L} = \sum_{n=1}^N C_n(q_n) + \lambda \left[\sum_{n=1}^N a_n(u_n - q_n) + B - \bar{K} \right]$$

First-order conditions

Differentiating with respect to each q_n ($n = 1, \dots, N$):

$$\frac{\partial \mathcal{L}}{\partial q_n} = \frac{\partial C_n}{\partial q_n} - \lambda a_n = 0 \quad \implies \quad \frac{MAC_n}{a_n} = \lambda^* \quad \forall n$$

Interpretation: at the optimum, every source equates its **marginal abatement cost per unit of concentration reduced** to the common shadow price λ^* .

The mathematics of cost-effective pollution control (4/9)

Case 1 — Uniformly mixed pollutants ($a_n = 1 \ \forall n$)

When every unit of emission has the **same** impact on the receptor ($a_n = 1$), the general FOC simplifies to:

$$MAC_n = \lambda^* \quad \forall n$$

This is the **equimarginal principle**: at the cost-effective allocation, all sources have **identical** marginal abatement costs.

Why emission standards fail

A uniform emission standard assigns a fixed abatement level $q_n = \bar{q}$ to every source. Because cost functions differ across firms, $MAC_n(\bar{q})$ will generally **not** be equal across sources:

$$MAC_i(\bar{q}) \neq MAC_j(\bar{q}) \quad \implies \quad \text{not cost-effective}$$

An individualised standard *could* work, but would require the regulator to know every firm's $C_n(\cdot)$ — information that firms have no incentive to reveal truthfully.

The mathematics of cost-effective pollution control (5/9)

Case 1 (cont.) — Emission charge

The regulator levies a per-unit charge T on every unit of pollution emitted. Each profit-maximising firm n solves:

$$\min_{q_n} C_n(q_n) + T(u_n - q_n)$$

- $C_n(q_n)$: cost of abating q_n units.
- $T(u_n - q_n)$: tax bill on remaining emissions.

First-order condition:

$$\frac{\partial C_n}{\partial q_n} = T \quad \implies \quad MAC_n = T \quad \forall n$$

Intuition: the firm abates while $MAC < T$ (cheaper than paying the tax), stops where $MAC = T$, and **pays the tax** on whatever it still emits. Because every firm faces the **same** T , they independently choose $MAC_1 = \dots = MAC_N = T$. If $T = \lambda^*$, this replicates the cost-effective allocation **exactly**. The regulator finds λ^* by **iterating**: raise T if total emissions exceed the target; lower T if they undershoot.

The mathematics of cost-effective pollution control (6/9)

Case 1 (cont.) — Cap-and-trade

The regulator issues $\bar{E}$ tradeable allowances (one per unit of permitted emission). Source n holds Ω_n allowances and can buy or sell at market price P . Each firm solves:

$$\min_{q_n} \underbrace{C_n(q_n)}_{\text{abatement cost}} + P \left[\underbrace{(u_n - q_n)}_{\text{actual emissions}} - \underbrace{\Omega_n}_{\text{permits held}} \right]$$

The bracketed term is the firm's **net permit position**: positive $\Rightarrow$ must **buy**; negative $\Rightarrow$ can **sell**.

First-order condition:

$$\frac{\partial C_n}{\partial q_n} = P \quad \Rightarrow \quad MAC_n = P \quad \forall n$$

Market clearing ($\sum \Omega_n = \bar{E}$) pins down the equilibrium price $P^* = \lambda^*$, the **same** shadow price as the regulator's problem, discovered by the market **without** trial and error.

Whether allowances are **gifted** or **auctioned**, the equilibrium price and abatement allocation are identical; only the **distribution of rents** differs.

The mathematics of cost-effective pollution control (7/9)

Case 2 — Non-uniformly mixed: ambient charge

When transfer coefficients **differ** ($a_n \neq a_m$), a **uniform** emission charge sets $MAC_n = T \ \forall n$, but the cost-effective condition requires $MAC_n/a_n = \lambda^*$. A uniform charge **ignores location** and is not cost-effective.

Instead, the regulator sets a **source-specific** ambient charge $t_n = a_n F$, where F is a common fee per unit of concentration. Each firm solves:

$$\min_{q_n} C_n(q_n) + a_n F(u_n - q_n)$$

First-order condition:

$$\frac{\partial C_n}{\partial q_n} = a_n F \quad \implies \quad \frac{MAC_n}{a_n} = F \quad \forall n$$

Set $F = \lambda^*$ and the cost-effective allocation is achieved. Sources **closer** to the receptor (a_n large) face a **higher** charge and abate more.

The mathematics of cost-effective pollution control (8/9)

Case 2 (cont.) — Non-uniformly mixed: ambient permits

Each permit authorises a concentration increase of ΔK_R at the receptor. Because source n raises concentration by a_n per unit emitted, it needs:

$$\underbrace{\frac{a_n}{\Delta K_R}}_{\text{permits per unit of emission}}$$

- A source **close** to the receptor (a_n large) needs **more** permits per tonne emitted.
- A source **far** away (a_n small) needs **fewer** permits per tonne.

The market-clearing permit price is $P^* = \lambda^*$. Each firm sets:

$$MAC_n = a_n P^* \quad \implies \quad \frac{MAC_n}{a_n} = \lambda^* \quad \forall n$$

Close sources face a **higher effective price** ($a_n P^*$) and abate more, exactly the cost-effective allocation.

The mathematics of cost-effective pollution control (9/9)

Case 2 (cont.) — The trading ratio

When source j (far) sells permits and source i (close) buys, the trade must keep concentration at R unchanged:

$$\underbrace{a_i \Delta E_i}_{\text{concentration rise from buyer}} = \underbrace{a_j \Delta E_j}_{\text{concentration fall from seller}}$$

Rearranging:

$$\Delta E_i = \frac{a_j}{a_i} \Delta E_j \quad (\text{trading ratio})$$

- If $a_i > a_j$ (buyer closer to receptor): $\Delta E_i < \Delta E_j$ — the buyer gets to emit **less** than the seller gave up.
- Total emissions may **fall**, but concentration at R stays unchanged — this is by design.

Appendix

References

- Goulder, L.H. (2013), "Markets for Pollution Allowances: What Are the (New) Lessons?" *Journal of Economic Perspectives* 27(1), 87–102.
- Hanley, N., Shogren, J.F. & White, B. (2019), *Environmental Economics in Theory and Practice*, 3rd ed. (Red Globe Press), Ch. 5.
- Kolstad, C.D. (2011), *Environmental Economics*, 2nd ed. (Oxford University Press), Ch. 11.
- Perman, R., Ma, Y., Common, M., Maddison, D. & McGilvray, J. (2011), *Natural Resource and Environmental Economics*, 4th ed. (Pearson), Ch. 7.
- Schmalensee, R. & Stavins, R.N. (2017), "Lessons Learned from Three Decades of Experience with Cap and Trade," *Review of Environmental Economics and Policy* 11(1), 59–79.
- Weitzman, M.L. (1974), "Prices vs. Quantities," *Review of Economic Studies* 41(4), 477–491.